

Analytical problem solving:-1. Perception Model:-

It is used for binary classification, which means it classifies the input into one of two categories.

Given:-

$$\text{Weight for Study Hours } (w_1) = 0.6$$

$$\text{Weight for Attendance } (w_2) = 0.4$$

$$\text{Bias } (b) = -0.5$$

$$\text{Study Hours } (x_1) = 2$$

$$\text{Study Hours } (x_2) = 1$$

Step-1 :- Calculate Net Input

$$\text{Net Input} = (w_1 \times x_1) + (w_2 \times x_2) + b$$

$$= (0.6 \times 2) + ((0.4) \times 1) + (-0.5)$$

$$= 1.2 + 0.4 - 0.5 = 1.1$$

$$\therefore \text{Net Input} = 1.1$$

Step-2 :- Apply step Activation function

The step Activation function is:

* If Net Input $\geq 0 \rightarrow \text{output} = 1$

* If Net output $< 0 \rightarrow \text{output} = 0$

$$\therefore \text{Since } 1.1 > 0$$

$$\text{output} = 1$$

$\therefore$ Since the output is 1, the perception predicts that the student passes.

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2. Backpropagation/ Delta Learning Rules:-

Backpropagation is a learning algorithm used in neural networks to reduce prediction errors.

Given:-

$$\text{Initial weight}(w) = 0.5$$

$$\text{Learning rate}(\eta) = 0.1$$

$$\text{Input}(x) = 2$$

$$\text{Actual output} = 0.6$$

$$\text{Target output} = 1$$

Step-1:- calculate Error

$$\boxed{\text{Error} = \text{Target} - \text{Actual}}$$

$$= 1 - 0.6 = 0.4$$

$$\text{Error} = 0.4$$

Step-2:- calculate weight update

$$\boxed{\Delta w = \eta \times \text{Error} \times \text{Input}}$$

$$= 0.1 \times 0.4 \times 2 = 0.08$$

$$\text{weight update} = 0.08$$

Step-3:- calculate New weight

$$\boxed{\text{New weight} = \text{old weight} + \Delta w}$$

$$= 0.5 + 0.08 = 0.58$$

$$\text{New weight} = 0.58$$

$\therefore$ The neuron's weight is increased from 0.5 to 0.58, helping the neural network reduce the prediction error and improve its learning.

Confusion Matrix Evaluation:-

It is a table used to evaluate the performance of a classification model. It compares the model's predicted values with actual values.

Given:-

$$\text{True Positive (TP)} = 180$$

$$\text{True Negative (TN)} = 150$$

$$\text{False Positive (FP)} = 20$$

$$\text{False Negative (FN)} = 50$$

Step-1:- calculate Accuracy

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \times 100$$

$$= \frac{180 + 150}{180 + 150 + 20 + 50} \times 100$$

$$= \frac{330}{400} \times 100 = 82.5\%$$

Step-2:- calculate precision

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \times 100$$

$$= \frac{180}{180 + 20} \times 100 = \frac{180}{200} \times 100$$

$$= 90\%$$

Step-3:- calculate Recall

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \times 100$$

$$= \frac{180}{180 + 50} \times 100 = \frac{180}{230} \times 100 = 78.26\%$$

Step-4: calculate Specificity

$$\text{specificity} = \frac{TN}{TN+FP} \times 100$$

$$= \frac{150}{150+20} \times 100 = \frac{150}{170} \times 100$$

$$= 88.24\%$$

∴ The model has good accuracy and precision.

4. Genetic Algorithm:-

It is an optimization technique inspired by the process of natural selection. It searches for the best solution using selection, crossover, mutation.

Given:-

fitness function $f(x) = x^2$

Initial population:- $\{2, 4, 6, 8\}$

Step-1: calculate fitness values

Fitness means how good each solution is.

Individual	fitness calculation	fitness value
2	2^2	4
4	4^2	16
6	6^2	36
8	8^2	64

Step-2: select the Best Individuals

choose individuals with the highest fitness values.

The individuals are 8 and 6 are having highest fitness value.

Step-3:- Perform crossover

Convert into binary:-

$$8 = 1000 \text{ and } 6 = 0110.$$

Exchange the least significant (last) bit.

Before crossover:- 1000 and 0110.

The last bits are both '0', so after crossover it remain the same.

$$1000 = 8 \text{ and } 0110 = 6.$$

Step-4:- Mutation

Mutation Means changing one bit randomly.

Ex:- $1000 \rightarrow 1001$.

∴ The genetic algorithm selects the individuals with the highest fitness.

5. Deep Neural Network performance Analysis :-

It is a type of artificial neural network with multiple hidden layers.

* Training Accuracy.

* Validation Accuracy.

Given:-

During training:-

→ Training Accuracy increases from 60% to 95%.

→ Validation Accuracy increases from 58% to 72% initially.

→ Later, validation Accuracy decreases to 68%.

Step-1:- Analyze the Accuracy Trend

Initially:-

- Training Accuracy increases from 60% to 95%.
- validation Accuracy also increases from 58% to 72%.

Later:-

- Training Accuracy continues to increase.
- validation Accuracy decreases from 72% to 68%.

Step-2:- Identify the issue

The issue is overfitting.

⇒ overfitting occurs when the model learns the training data too well.

Techniques:-

- * Early stopping.
- * Regularization.

Importance of validation Accuracy.

- * Measures the performance of the model on unseen data.
- * Helps to detect overfitting.
- * Helps to compare different models.